Exploratory Data Analysis on the Automobile Dataset

Report

Introduction

This analysis utilises the automobile.txt dataset, which contains technical specifications and market information for various vehicles. The data includes features such as manufacturer (make), fuel type, engine size, horsepower, fuel efficiency (MPG), and price. The primary goal of this EDA is to identify key relationships between vehicle performance measures and market value.

Data cleaning

The dataset was loaded into a pandas DataFrame. The redundant normalized-losses and symboling columns were removed. No duplicate rows were found. Question marks were treated as missing values, and 12 incomplete rows were removed, leaving 193 complete records. The price, horsepower, and peak-rpm columns were then converted to the NumPy int64 data type for analysis.

Missing data

Some records contained missing values, represented by question marks. These entries were removed after being identified as missing data so that the calculations and visualisations use only complete vehicle records.

Data stories and visualisations

- Price-Efficiency: The five most expensive cars average 15.2 city MPG, compared with 36.0 city MPG for the five cheapest cars. Across the complete dataset, price and city MPG have a negative relationship (Pearson r = -0.71).

- Manufacturer Specialisation: Chevrolet has the highest average city MPG in this dataset, at 41.0 MPG. This result is based on the vehicles represented in this dataset.

- Engine Size vs. Performance: The largest engine is 326 cubic inches (Jaguar). Engine size, horsepower, and price are positively related, while they are negatively related to fuel economy.

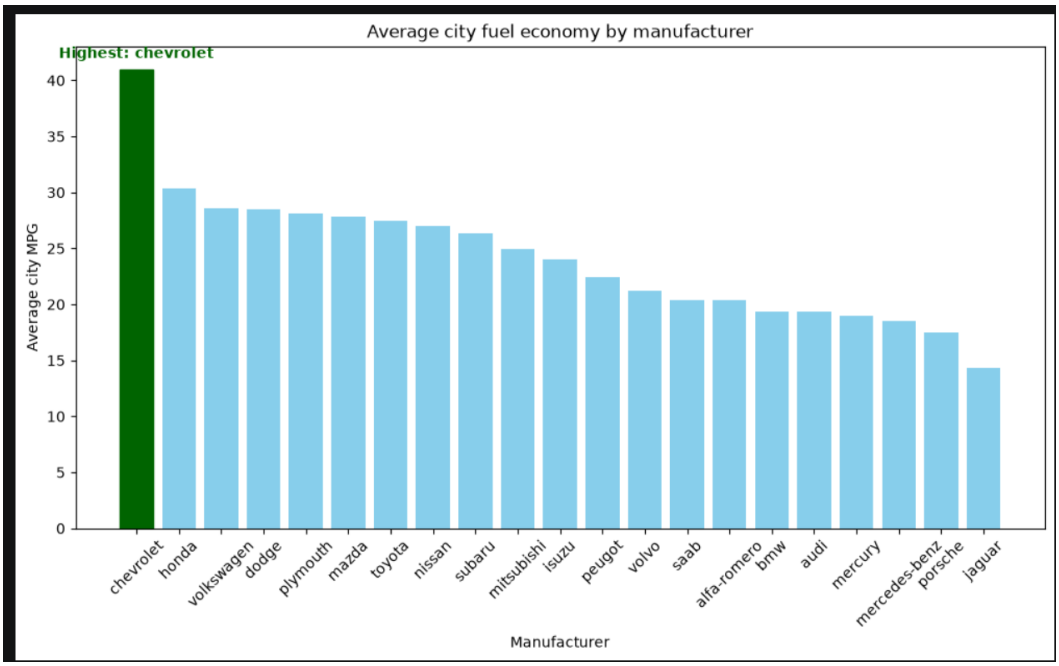


Figure 1. Average city MPG by manufacturer.

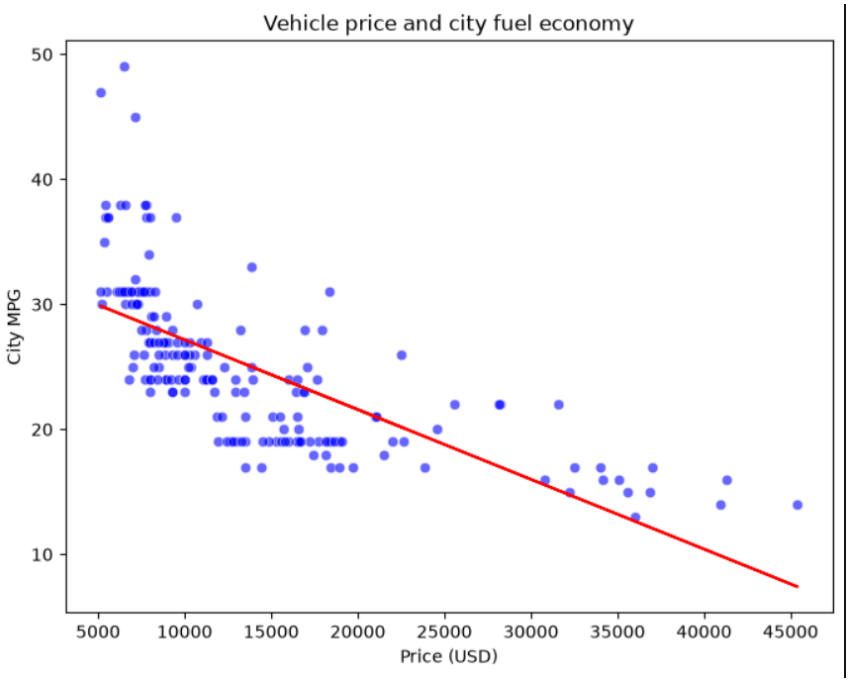


Figure 2. Higher-priced vehicles tend to have lower city MPG.

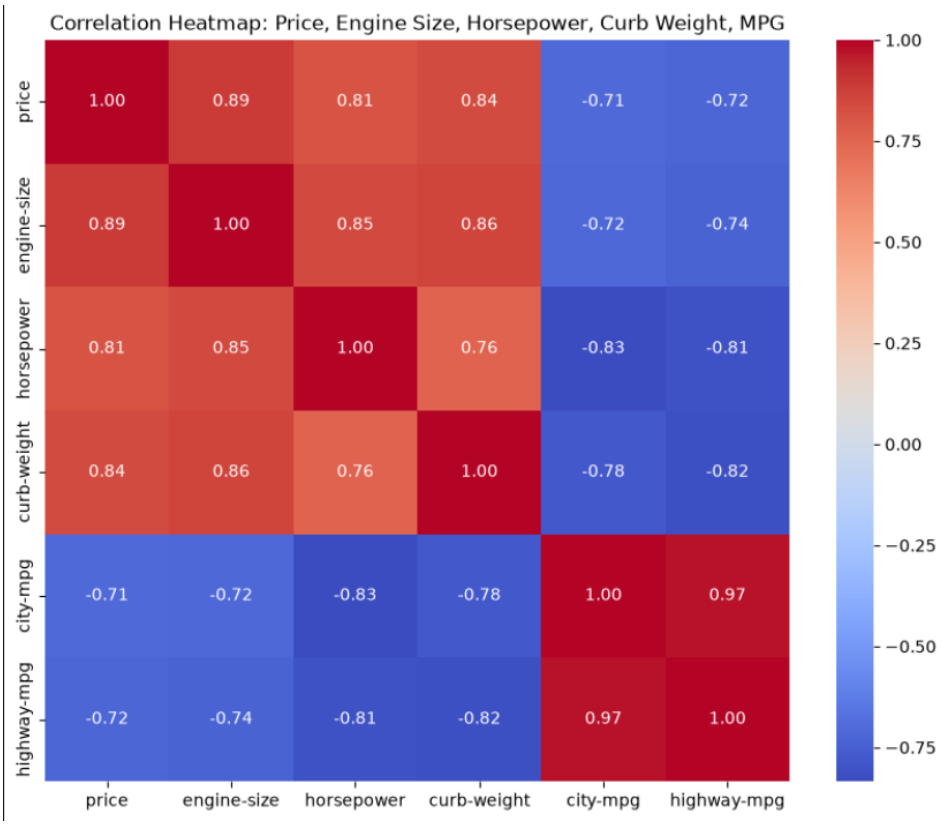


Figure 3. Correlations among price, engine size, performance, and MPG.

This report was written by: Lesedi Ramasoga